

EduGenie: Google Gemini Powered Learning Assistant

Phase 3

Project Workflow

MILESTONE 1: Model Selection and Architecture

Activity 1.1: Select AI Models

- Gemini 1.5 Pro (via API):

Used for: Q&A, summarization, quiz generation, learning paths

Benefits: Advanced reasoning, structured outputs, and cloud inference

- LaMini-Flan-T5-783M (local):

Used for: Concept explanation

Benefits: Instruction-tuned, lightweight, CPU-compatible

Folder Architecture:

EduGenie/

1. main.py # FastAPI app
2. explanation_module.py # Concept explanation logic
3. qna.py # Question answering
4. quiz_module.py # Quiz generation
5. summary_module.py # Summarization
6. learning_path.py # Learning recommendations
7. templates/index.html # HTML frontend
8. static/style.css # Styling
9. requirements.txt # Python dependencies

FIG. Showing the Folder Architecture